

3. Connect Live Meteorological Weather API

Target Goal: Feed real-time microclimate metrics (relative humidity, rainfall, dew-point, wind) directly into the disease spread algorithm using the free Open-Meteo API.

WHY WEATHER DRIVES DISEASE SPREAD

Fungal spores (such as *Alternaria solani* and *Phytophthora infestans*) cannot germinate on dry leaves. They require $\geq 80\%$ relative humidity or standing water droplets for 12 to 24 consecutive hours. Connecting hyper-local weather turns a static diagnosis into an early warning radar.

Step 1: The Open-Meteo API Endpoint

The Open-Meteo API is 100% free for commercial & non-commercial use, requires no API key, and offers 1km spatial resolution:

```
GET https://api.open-meteo.com/v1/forecast?latitude=22.7196&longitude=75.8577&hourly=temperature_2m,rel
```

Step 2: Frontend Weather Service (`src/services/weatherService.ts`)

```
// src/services/weatherService.ts
export interface LiveWeatherReport {
  temperature: number;
  humidity: number;
  rainProbability: number;
  windSpeed: number;
  fungalRiskLevel: 'Low' | 'Moderate' | 'Critical';
  advisory: string;
}

export async function fetchFieldWeather(lat: number, lng: number): Promise<LiveWeatherReport> {
  const url = `https://api.open-meteo.com/v1/forecast?latitude=${lat}&longitude=${lng}&hourly=temperature,relativehumidity,precipitationprobability,windspeed`;

  const res = await fetch(url);
  if (!res.ok) throw new Error('Weather API fetch failed');
  const data = await res.json();

  const currentTemp = data.current_weather.temperature;
  const currentWind = data.current_weather.windspeed;
  const currentHumidity = data.hourly.relativehumidity_2m[0];
  const currentRainProb = data.hourly.precipitation_probability[0] || 0;

  // Scientific Fungal Spore Germination Risk Logic
  let risk: 'Low' | 'Moderate' | 'Critical' = 'Low';
  let advisory = 'Environmental conditions are stable for crop growth.';

  if (currentHumidity ≥ 85 && currentRainProb ≥ 60) {
    risk = 'Critical';
    advisory = 'High humidity (>85%) and rain expected. Critical risk of fungal spore germination!';
  } else if (currentHumidity ≥ 75 || currentRainProb ≥ 40) {
    risk = 'Moderate';
    advisory = 'Elevated leaf moisture detected. Inspect lower canopy foliage.';
  }

  return {
    temperature: currentTemp,
    humidity: currentHumidity,
    rainProbability: currentRainProb,
    windSpeed: currentWind,
    fungalRiskLevel: risk,
    advisory
  };
}
```

Step 3: Calculating 7-Day Disease Spread Progression

Use this mathematical model to project how severity expands over 7 days based on current humidity:

```
// Calculates 7-day severity trajectory
export function calculateSpreadTrajectory(baseSeverity: number, humidity: number, rainProb: number) {
  // Humidity multiplier: >80% humidity accelerates growth by up to 1.8x
  const humidityFactor = humidity > 80 ? 1 + (humidity - 80) * 0.03 : 1.0;
  const rainFactor = 1 + (rainProb / 100) * 0.2;

  const trajectory = [];
  let current = baseSeverity;

  for (let day = 1; day ≤ 7; day++) {
    const dailyGrowth = current * 0.08 * humidityFactor * rainFactor;
    current = Math.min(95, Math.round(current + dailyGrowth));
    trajectory.push({ day: `Day ${day}`, severity: current });
  }

  return trajectory;
}
```